



## Electronic Engineering Internship at Xidian University, Xi'an



French engineering students are invited to participate **in an internship in Electronic Engineering at Xidian University, Xi'an**, one of China's leading universities in electronics, communications, and information technology.

During the internship, students will:

- Work on electronic circuits, embedded systems, and signal processing projects
- Gain hands-on experience with modern lab equipment and engineering tools
- Collaborate in a multicultural research environment with Chinese students and faculty

This internship offers a unique opportunity to enhance technical skills and explore China's innovation in electronic engineering.

**Subject :** Design and Implementation of Crop Recognition and Precision Picking System in Agricultural Scenarios Based on Mobile Picking Robot.

## **Type :**

Design Category (Integration of Autonomous Navigation in Agricultural Scenarios, Crop Recognition and Maturity Judgment, Precision Control of Robotic Arm, Focusing on Competition-Oriented and Practical Agricultural System Integration)

## **Task Description :**

This topic takes the "Future Agricultural Robot Challenge" of the iCAN College Student Innovation and Entrepreneurship Competition as the application background, relying on mobile picking robots (including Mecanum wheel chassis, 3-4 DOF robotic arm, depth camera and other equipment) to develop a comprehensive system integrating "autonomous navigation-crop recognition-precision picking-system optimization". It solves the problems of environmental adaptation, recognition accuracy and picking efficiency in agricultural picking scenarios. The specific tasks are as follows

### **1. Autonomous Navigation of Robot and Picking Path Planning in Agricultural Scenarios**

Based on the laser radar (detection radius  $\geq 15\text{m}$ , anti-ambient light  $\geq 80\text{K Lux}$ ) and IMU module (3-axis acceleration  $\pm 16\text{g}$ , angular velocity  $\pm 2000^\circ/\text{s}$ ) equipped on the robot, develop the SLAM mapping algorithm under the ROS system to construct a  $10\text{m} \times 5\text{m}$  simulated agricultural scene map (marking crop planting rows and irrigation facility obstacle areas); combined with the "zero spin radius" characteristic of the Mecanum wheel system (spin radius  $\leq 0\text{mm}$ ), design an A\* path planning algorithm suitable for crop row spacing, realize the robot's autonomous cruise along the planting row (maximum speed  $\geq 0.6\text{m/s}$ ), ultrasonic obstacle avoidance (detection range  $20\text{-}600\text{cm}$ , avoiding obstacles with diameter  $\geq 5\text{cm}$ ) and precise docking at picking points (positioning error  $\leq 2\text{cm}$ ), adapting to the narrow working space of agricultural scenarios such as greenhouses.

### **2. Development of Crop Recognition and Maturity Judgment Module**

Collect data of simulated crops commonly used in competitions (such as strawberries and tomatoes) based on a depth camera (RGB resolution  $1920 \times 1080@15\text{fps}$ , depth range  $0.2\text{-}3\text{m}$ ), and construct a crop image dataset with "unripe-ripe-overripe" labels (sample size  $\geq 1000$  images); use YOLOv5s/MobileNet lightweight deep learning models to develop crop recognition and maturity classification algorithms, improve agricultural scene adaptability through transfer learning (pre-trained based on COCO dataset), output 2D coordinates of RGB images and 3D depth positions of mature crops, ensuring recognition accuracy  $\geq 92\%$  and single-frame processing time  $\leq 0.3\text{s}$ ; combined with the  $60^\circ$  field of view of the camera module, optimize the image acquisition angle ( $30^\circ\text{-}45^\circ$  with the crop row) and frequency ( $10\text{fps}$ ) to reduce missed recognition caused by leaf occlusion.

### **3. Precision Picking Control of Robotic Arm and Adaptive Gripper**

Complete the hand-eye calibration of the 3-4 DOF robotic arm (repeat positioning accuracy  $\pm 0.1\text{mm}$ , working radius  $\geq 340\text{mm}$ ) based on ROS hand-eye calibration tools, establish a coordinate mapping model of "depth camera-robotic arm base coordinate-gripper end", and convert the 3D position of crops into robotic arm motion commands; develop picking

trajectory planning logic through Python programming interface, control the robotic arm to act according to the process of "approaching crops (end speed  $\geq 100\text{mm/s}$ )  $\rightarrow$  adaptive gripper opening (clamping range 0-45mm)  $\rightarrow$  precise clamping (automatic force adjustment according to crop diameter)  $\rightarrow$  lifting off the plant  $\rightarrow$  transferring to the storage box"; optimize the gripper control strategy: adjust the clamping force (0-150g) through pressure sensor feedback for different crops such as strawberries (diameter 2-4cm) and tomatoes (4-6cm) to avoid crop damage or falling, ensuring a picking success rate  $\geq 88\%$ .

#### 4. System Integration and Performance Optimization

Integrate the voice control function (based on TTL serial voice module, 5V power supply), support voice commands such as "start picking", "pause picking" and "switch crop type" (response time  $\leq 1\text{s}$ ) to adapt to the convenient operation needs of agricultural scenarios; based on the 12th generation N100 quad-core controller (TDP 6W, cache 6MB), use multi-threading technology to optimize data transmission between navigation, recognition and picking modules, control the delay between modules  $\leq 0.5\text{s}$ , and increase the picking volume per unit time to  $\geq 15$  pieces/minute; build a simulated agricultural scene test platform, monitor the stability of the robotic arm motor PWM signal (ripple  $\leq 100\text{mV}$ ) and system power consumption (full-load cruise  $\leq 15\text{W}$ ) with oscilloscopes and multimeters, optimize the lithium

battery power supply strategy (working time  $\geq 4\text{h}$ ), and test the system stability at 0~40°C and 5%-80% humidity to ensure trouble-free operation  $\geq 8\text{h}$